Module 3 Homework 1 Challenge

Below is a scored review of your assessment. All questions are shown.

Earned 79,992 of 99,990 points. **80%**

Main Question Set

Solve six of the first ten questions to achieve a passing score of 60%. Don't miss the learning opportunity in the rest of the problems, though -- spend the full time and attempt every one of them!

Correct Answer Partially Correct Incorrect Answer

1

Find the value of the next term in this sequence:

$$1, 1, 2, 3, 5, 8, 13, 21, ?$$

Your Answer	34
Correct Answer	34

Explanation

This is the Fibonacci sequence that you saw in the Day 4 video! We can write the value of the nth term as $f(n) = f(n-1) + f(n-2)$. Then the next term in the sequence will be $21 + 13 = 34$.

2 Find which of the following statements is true:

Your Answer	314×316 ends in 24
Correct Answer	314×316 ends in 24

Explanation

You could multiply all of these numbers out, but remember what the video lesson from Day 2 showed us? Let's take a look at choice three, 314×316.

We can write this as

$$\left(310+4\right)\left(310+6\right) = 310^2 + \left(4+6\right)310 + 4 \times 6$$

The terms that could normally contribute to the last two digits would be the $(4+6)310$ term and the 4×6 term, since the 310^2 term ends in 00. However, since $4+6$ adds to 10, and the 310 is naturally a multiple of 10, since it is the first part of the number, the first term is a multiple of 100. This means our last two digits come entirely from the 4×6.

Now, if the hundreds and tens digits weren't the same (like in our example, with the 31 in 314 and 316,) then we wouldn't be able to factor out $4+6$ from the cross term. So, in order for this trick to work, the digits of the number before the units digit must be the same, and the units digits must add to ten.

Our answer is choice number three.

3

The average of two numbers is a, and the two numbers can be written as $a + c$ and $a - c$. Then, the product of the two numbers is

Your Answer	$a^2 - c^2$
Correct Answer	$a^2 - c^2$

Explanation

The product of the two numbers is

$$\left(a + c\right)\left(a - c\right) = a^2 - c^2.$$

This is the Difference of Squares of two numbers, which as we can see can be nicely factored!

4 Suppose you have a two-digit number where integer a is the tens digit and integer b is the ones digit, and a and b are both single-digit numbers. Which of the following is an expression for the number that you get from swapping the digits of this number?

Your Answer	$10b + a$
Correct Answer	$10b + a$

Explanation

Our original number is equal to $10a + b$, so if you swap the digits, then b becomes the tens digit, and a becomes the ones digit. Then the value that b gives to the number is $10b$, so the number is $10b + a$.

5 The price of a donut started at $\$1$, and then went up by $x\%$. Later, the price fell by $x\%$. Now the price of the donut is 36% **less** than it was when it started at $\$1$. Find the percent x that it rose and fell by.

Your Answer	60%
Correct Answer	60%

Explanation

Did you notice the symmetry in this problem? The $x\%$ rise equals the $x\%$ drop in price. This is setting us up for a difference of squares problem!

After the price rose by $x\%$, the price of the donut became $\$1(1+x\%)$. After the price fell by $x\%$, the price of the donut became $1-x\%$ times the temporary increased price, so the total price after both changes is

$$\begin{aligned}
&\$1\left(1+x\%\right)\left(1-x\%\right)\\
&=\$1\left(1-x\%+x\%-(x\%)^2\right)\\
&=\$1\left(1-(x\%)^2\right)
\end{aligned}$$

Are you thinking that 36 is a square, which makes this really nice? Yes! The final price is

$$\begin{aligned}
\$1\left(1-(x\%)^2\right)&=\$1\left(1-\frac{36}{100}\right)\\
(x\%)^2&=\frac{36}{100}\\
x\%&=\sqrt{\frac{36}{100}}\\
x\%&=\frac{6}{10}\\
x\%&=\frac{60}{100}\\
x&=60
\end{aligned}$$

6 Let the operation @ be defined like this:

$$a@b = a - b$$

Find the value of

$$\Big((10@9)@(8@7)\Big)@(6@5)@\Big((4@3)@(2@1)\Big)$$

Your Answer -1

Correct Answer -1

Explanation

Let's start by calculating the innermost parentheses:

$$\begin{aligned}\Big((10@9)@(8@7)\Big)@(6@5)@\Big((4@3)@(2@1)\Big) &= \\ &= \Big(1@1\Big)@1@\Big(1@1\Big) \\ &= 0@1@0 \\ &= 0 - 1 - 0 \\ &= -1\end{aligned}$$

7

Let the function

$f(x) = 3x + 6$, if $x < 5$

$f(x) = 7x - 20$, if $x \geq 5$

Find the value of $f(f(f(2)))$.

✓	**Your Answer**	428
	Correct Answer	428

Explanation

Plugging in 2 to $f(x)$, we get $f(2) = 12$.

Plugging this into $f(x)$ again, we get $f(f(2)) = f(12) = 64$.

Plugging this into $f(x)$ again, we get $f(f(f(2))) = f(64) = 428$.

8 How many terms are in the expansion of $2(4x-4)(4x+4)$ when it is simplified to lowest terms?

Your Answer	2
Correct Answer	2

Explanation

Using Difference of Squares,

$$(4x-4)(4x+4) = 16x^2 - 16,$$

so 2 times this equals

$$32x^2 - 32.$$

Therefore, the number of terms is 2.

9 Find the value of x that makes the following true:

$$8^x \times 9^x = 19^x$$

✕	**Your Answer**	[no answer given]
	Correct Answer	0

Explanation

Since the terms are all put to the same power of x, we can combine them into one term by dividing:

$$\begin{aligned} 8^x \times 9^x &= 19^x \\ \frac{8^x \times 9^x}{19^x} &= 1 \\ \left(\frac{8 \times 9}{19}\right)^x &= 1 \end{aligned}$$

The only way for an exponent to equal 1 is for the power to equal 0, so $x = 0$.

10 Before you take your weekly Daily Challenge exam, which also has 20 questions, your teacher offers to pay you 7 cents for each correct answer you put, minus 3 cents for each incorrect answer you put. In the end, however, she tells you that she owes you zero cents. How many questions did you answer correctly?

✕	**Your Answer**	[no answer given]
	Correct Answer	6

Explanation

We need a variable, so how about letting the number of correct questions equal x? Then we also know the number of incorrect questions you answered, since that is just the total minus the number correct, or $20 - x$. The money you earn from the correct questions is $7x$ and the money your teacher subtracts from that amount is $3(20 - x)$. These two have to be equal to each other, so we have

$$7x = 3\left(20 - x\right).$$

$$7x = 60 - 3x$$

$$10x = 60$$

$$x = 6$$

The number of questions that you answered correctly, then, is 6, which of course isn't this exam that you are currently taking, since you are aceing this one!

Earned 0 of 7 points. 0%

Bonus Challenge Questions

These optional questions will not count towards your final score. However, we strongly suggest trying them for the benefit of the challenge!

11 If $\frac{2}{3}$ of the students in a class take Spanish, $\frac{1}{2}$ of the students in the class take French, and 6 students take both, then if everyone in the class takes at least one of either Spanish or French, how many total students are in the class?

Your Answer	[no answer given]
Correct Answer	36

Explanation

The total fraction of students who take both languages is

$$\frac{2}{3} + \frac{1}{2} - 1 = \frac{1}{6}$$

Since $\frac{1}{6}$ of the students in the class equals 6 students, the total number of students in the class is $6 \times 6 = 36$ students.

12 Which 3-digit integer starting with 6 has this property: that if you delete the first digit, the number formed from the remaining digits is 25 times smaller than the original number?

Your Answer	[no answer given]
Correct Answer	625

Explanation

We can write our three-digit number as $6 \times 10^2 + a$, where a is an integer. Then removing the first digit gives a remaining, and since a is 25 times smaller than the original number, then

$$\frac{6 \times 10^2 + a}{25} = a$$

$$6 \times 10^2 + a = 25a$$

$$6 \times 10^2 = 24a$$

$$a = \frac{600}{24}$$

$$= 25$$

which means that our number is

$$6 \times 10^2 + 25 = 625.$$

13

Find the value of x if $\frac{13x+11}{x+5} = 7$.

✕	**Your Answer**	[no answer given]
	Correct Answer	4

Explanation

Let's do operations to both sides of the equation until it looks better. First, let's get rid of the fraction by multiplying both sides by $x + 5$. Then we can multiply everything out (expand) and combine the terms to get x on one side, and numbers on the other side.

$$\begin{aligned} \frac{13x+11}{x+5} &= 7 \\ 13x + 11 &= 7(x+5) \\ 13x + 11 &= 7x + 35 \\ 6x &= 24 \\ x &= 4 \end{aligned}$$

14 Alice's brother Bob is twice the age of Alice. Her other brother Charlie is half the age of Alice. In 6 years, Charlie will be half the age of Bob. How old is Alice today?

Your Answer	[no answer given]
Correct Answer	6

Explanation

Let's denote Alice, Bob, and Charlie's present day ages as a, b, and c respectively, and try to convert each statement provided into algebraic equations. The first fact we get is that Bob is twice the age of Alice, so we get the relation

$$b = 2a.$$

The second statement states that Charlie is half the age of Alice, so we get a similar equation,

$$c = \frac{1}{2}a,$$

or

$$2c = a.$$

With these facts, we can plug the second equation into the first to get

$$b = 2(2c) = 4c.$$

The final bit of information we are given is the fact that Charlie will be half the age of Bob in 6 years. At that point in time, Charlie will be $c + 6$ years old and Bob will be $b + 6$ years old, so we get the equation

$$(c + 6) = \frac{1}{2}(b + 6),$$

which again can be simplified to

$$2(c + 6) = (b + 6).$$

Plug in

$$b = 2(2c) = 4c,$$

and we get

$$2(c + 6) = (4c + 6),$$

which further simplifies to $c = 3$, so we know that Charlie is 3 years old today. Hence, Alice must be $2(3) = 6$ years old.

15 Starting from 1000, Tim divides the number that he currently has by 2 and rounds down. Find the number of times he does this before reaching the number 1.

Your Answer	[no answer given]
Correct Answer	9

Explanation

This problem is similar to the Day 5 video problem that showed an arithmetic sum made up of powers of 2. $(512 + 256 + 128 + 64 + 32 + 16 + 8 + 4 + 2 + 1)$. In this problem, we are approximating the terms based on powers of 2 as well, and we want to know how many terms, so that is like figuring out which 2^n is closest to our starting number, 1000.
The closest power of 2 to 1000 is 2^{10}, which equals 1024. Since 1000 is smaller than 1024, we have one less term than 10, so the answer is 9.

16 Find the largest 2-digit number that has this property: if the last digit is deleted, then the number is divisible by the new number.

Your Answer	[no answer given]
Correct Answer	99

Explanation

Variables to the rescue! Let's call our 2-digit number ab, so the number that you get after deleting the last digit is a. Since b can be divided by a, then there is some integer n that gives

$$10a + b = n \times a.$$

It wouldn't make sense if n were smaller than 10. We intuitively know that numbers that are a multiple of 10 already work in this case. For example, $160 = 10 \times 16$. If n were smaller than 10, then the first digits of b and a couldn't be the same.

The multiplicative factor could be greater than 10, since otherwise 10 would be our answer, and that's probably not the biggest possible b. Let's isolate the 10 from the factor n:

$$n = 10 + \overline{n}$$

Now, joining the two equations gives

$$10a + b = \left(10 + \overline{n}\right)a$$

$$10a + b = 10a + \overline{n}a$$

$$b = \overline{n}a$$

Remembering that b is a digit, this tells us that $\overline{n}$ is also a single digit, so $\overline{n} \leq 9$ and $n \leq 19$. The range for our multiplicative factor, n, is from 10 to 19.

We can use $b = \overline{n}a$ to find all the possible two-digit numbers:

$$\overline{n} = 1;\ b = a : 11, 22, 33, 44, 55, 66, 77, 88, 99$$
$$\overline{n} = 2;\ b = 2a : 12; 24; 48$$
$$\overline{n} = 3;\ b = 3a : 13; 26; 39$$
$$\overline{n} = 4;\ b = 4a : 14; 28$$
$$\overline{n} = 5;\ b = 5a : 15$$
$$\overline{n} = 6;\ b = 6a : 16$$
$$\overline{n} = 7;\ b = 7a : 17$$
$$\overline{n} = 8;\ b = 8a : 18$$
$$\overline{n} = 9;\ b = 9a : 19$$

From looking at these possibilities, the largest two-digit number is 99.

17 A rich merchant has many gold coins and doesn't want anybody to know about them. One day his wife asked, "How many gold coins do you have?" The merchant replied, "If I divide the coins into two unequal numbers, then 36 times the difference between the two numbers equals the difference between the squares of the two numbers." How many gold coins does the merchant have?

Your Answer	[no answer given]
Correct Answer	36

Explanation

Let the two unequal non-negative integers be x and y. One of them must be greater than the other, so let $x > y$. This is so we don't have to deal with negatives after the difference. We are given $36(x-y) = x^2 - y^2$. Factoring the right hand side, we obtain $36(x-y) = (x+y)(x-y)$. Since $x \neq y$, we can divide both sides by $(x-y)$ and not have to worry about dividing by 0. This gives $36 = x + y$.

Earned 0 of 3 points. 0%

Super Bonus Questions

These questions are an extreme challenge! Some of the most interesting, creative and clever questions are the hardest ones. So try every single one. Even if you can't figure it out, you'll still be deepening your understanding of the important math concepts that underlie this course.

18 A dishwasher that costs $\$3600$ normally is on sale for 35% off. An oven that normally costs $\$10,400$ is now on sale for 15% off. If I buy both the dishwasher and the oven at their discounted prices, what fraction of the normal price total am I paying? Your answer should be a fraction without spaces, e.g. 1/2

Your Answer	[no answer given]
Correct Answer	559/700

Explanation

The fraction of the normal price that you pay is $\frac{\text{price}-\text{discount}}{\text{price}}$. It might be easier to find just $\frac{\text{discount}}{\text{price}}$ and then find $1-\frac{\text{discount}}{\text{price}}$. Let's find the value of the top and the bottom. In order to find the top, we need to find the total discount.

$$\begin{aligned}\text{dishwasher discount} &= 35\% \times \$3600\\ &= \frac{35}{100} \times \$3600\\ &= 35 \times \$36\\ &= \$1260\end{aligned}$$

$$\begin{aligned}\text{oven discount} &= 15\% \times \$10,400\\ &= \frac{15}{100} \times \$10,400\\ &= 15 \times 104\\ &= \$1560\end{aligned}$$

So the total discount is $\$1260 + \$1560 = \$2820$. The total normal price is $\$3600 + \$10,400 = \$14,000$.

The fraction $\frac{\text{discount}}{\text{price}} = \frac{\$2820}{\$14,000} = \frac{282}{1400} = \frac{141}{700}$. We want

$$1 - \frac{141}{700} = \frac{559}{700}.$$

19 Two numbers are in the ratio $1:3$. The first number starts with 1, and the second number is produced by altering the first number in this way: the 1 at the start is removed and placed at the end. What is the smallest possible value of the smaller number, which starts with 1?

✕	**Your Answer**	[no answer given]
	Correct Answer	142857

Explanation

Let's try to quantify the meaning of the relationship between the two numbers, using math. Since the 1 at the beginning of a number represents a power of 10, then our first number can be written as

$$10^n + a$$

where a is the number that we get after deleting the 1 in front, without adding it to the end. We don't know how many digits this number has yet, but that's okay; here it has $n+1$ digits.

Our second number has the same digits as a, but shifted to the left by one spot, which means that they all are multiplied by 10. Appending 1 to the end means that the ones digit of our larger number is 1. Using the information that these two numbers are in the ratio $1:3$, we know the larger is three times as big as the smaller, and we can write

$$\left(10^n + a\right)3 = 10a + 1$$

$$3 \times 10^n + 3a = 10a + 1$$

$$3 \times 10^n - 1 = 7a$$

$$a = \frac{3 \times 10^n - 1}{7}$$

We know that a is an integer, so $3 \times 10^n - 1$ must be a multiple of 7. These numbers have the form $299999..999$, and we can try dividing by 7 to look for the smallest number that works. This happens to be $299,999$. Then $a = 42,857$, and that means that our smaller number is

$$10^5 + 42,857$$

$$= 142,857.$$

20 Find the value of

$$\frac{1}{2}+\frac{2}{2^2}+\frac{3}{2^3}+\frac{4}{2^4}+\frac{5}{2^5}+\ldots$$

Your Answer [no answer given]

Correct Answer 2

Explanation

Let's make a variable for the quantity that we want to find, and call it x. Now, let's divide x by 2:

$$\begin{aligned}\frac{x}{2}&=\frac{1}{2}\left(\frac{1}{2}\right)+\frac{1}{2}\left(\frac{2}{2^2}\right)+\frac{1}{2}\left(\frac{3}{2^3}\right)+\frac{1}{2}\left(\frac{4}{2^4}\right)+\frac{1}{2}\left(\frac{5}{2^5}\right)+\ldots\\&=\frac{1}{2^2}+\frac{2}{2^3}+\frac{3}{2^4}+\frac{4}{2^5}+\frac{5}{2^6}+\ldots\end{aligned}$$

This looks like x, except that we are missing the $\frac{1}{2}$, and the numerators are off by 1. If we subtract the expression we got from x, we get

$$\begin{aligned}&\frac{1}{2}+\frac{2}{2^2}+\frac{3}{2^3}+\frac{4}{2^4}+\frac{5}{2^5}+-\left(\frac{1}{2^2}+\frac{2}{2^3}+\frac{3}{2^4}+\frac{4}{2^5}+\frac{5}{2^6}+\ldots\right)\\&=\frac{1}{2}+\frac{1}{2^2}+\frac{1}{2^3}+\frac{1}{2^4}+\ldots\\&=x-\frac{x}{2}\\&=\frac{x}{2}.\end{aligned}$$

We did this because it's possible to find the value of this series, since it is a normal geometric series. To avoid confusion, let's call this series by a difference name, like g :

$$\begin{aligned}g&=\frac{1}{2}+\frac{1}{2^2}+\frac{1}{2^3}+\ldots\\g-\frac{1}{2}&=\frac{1}{2^2}+\frac{1}{2^3}+\ldots\\g-\frac{1}{2}&=\frac{g}{2}\\\frac{g}{2}&=\frac{1}{2}\\g&=1\end{aligned}$$

Remember what we're trying to find? It's not g, but x, where

$$\frac{x}{2} = \frac{1}{2} + \frac{1}{2^2} + \frac{1}{2^3} + \ldots$$
$$\frac{x}{2} = g$$
$$\frac{x}{2} = 1$$
$$x = 2$$

So, to answer our question, the sum of this series is 2.

Summary

Software by

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